Experiment No. 2

Aim: To convert the ER diagram of an into a relational schema. Write complete MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCODE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Relational Schema

CUSTOMER (
    CustomerID PK,
    FirstName,
    Last Name,
    Email UNIQUE,
    DOB,
    Gender,
    Phone No,
    CreatedAt
)

ADDRESS (
    AddressID PK,
    CustomerID FK,
    Address Type,
    Is Default,
    Street,
    City,
    State,
    Pincode
)

SELLER (
    SellerID PK,
    Seller Name,
    Phone,
    Email UNIQUE,
    GSTIN UNIQUE
)

CATEGORY (
    CategoryID PK,
    Parent CategoryID FK,
)

PRODUCT (
    Product ID PK,
    Seller ID FK,
    CategoryID FK,
    Product Name,
    Price,
    Stock Qty,
    Description
)

ORDERS (
    Order ID PK,
    Customer ID FK,
    Order Date,
    Total Amt,
    Order Status
)

ORDER_ITEM (
    Order ID PK/FK,
    ItemSeq PK,
    Product ID FK,
    Address ID FK,
    Quantity,
    Unit Price,
    Subtotal
)

PAYMENT (
    PID PK,
    Order ID FK,
    ItemSeq FK,
    PMode,
    Amt,
    PStatus,
    PDate
)

DELIVERY (
    DID PK,
    OrderID FK,
    Item Seq FK,
    Courier Name,
    Status,
    Est Date,
    DeliveryDate
)

Source Code

```
-- EXPERIMENT 2
-- ER Diagram to Relational Schema
DROP DATABASE IF EXISTS dbms_lab;
CREATE DATABASE dbms_lab;
USE dbms_lab;
-- 1. CUSTOMER TABLE
CREATE TABLE Customer (
    CustomerID VARCHAR(10) PRIMARY KEY,
    FirstName VARCHAR(30) Not NULL,
    LastName VARCHAR(30) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    DOB DATE,
    Gender VARCHAR(10),
    PhoneNo VARCHAR(15),
    CreatedAt DATETIME NOT NULL DEFAULT CURRENT_TIMESTAP
);
-- 2. ADDRESS TABLE
CREATE TABLE Address (
    AddressID VARCHAR(10) PRIMARY KEY,
    CustomerID VARCHAR(10) NOT NULL,
    IsDefault BOOLEAN NOT NULL DEFAULT FALSE,
    Street VARCHAR(100) NOT NULL,
    City VARCHAR(50) NOT NULL,
    State VARCHAR(50) NOT NULL,
    Pincode VARCHAR(10) NOT NULL,

    CONSTRAINT fk_address_customer
        FOREIGN KEY (CustomerID)
        REFERANCES Customer (CustomerID)
        ON DELETE CASCADE
);
```

```
-- 3. SELLER TABLE
CREATE TABLE Seller (
    SellerID VARCHAR(10) PRIMARY KEY,
    SellerName VARCHAR(100) NOT NULL,
    Phone VARCHAR(15) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    GSTIN VARCHAR(20) NOT NULL UNIQUE
);

-- 4. CATEGORY TABLE
CREATE TABLE Category (
    CategoryID VARCHAR(10) PRIMARY KEY,
    ParentCategoryID VARCHAR(10) NULL,

    CONSTRAINT fk_category_parent
        FOREIGN KEY (ParentCategoryID)
        REFERENCES Category(CategoryID)
        ON DELETE SET NULL
);

-- 5. PRODUCT TABLE
CREATE TABLE Product (
    ProductID VARCHAR(10) PRIMARY KEY,
    SellerID VARCHAR(10) NOT NULL,
    CategoryID VARCHAR(10) NOT NULL,
    ProductName VARCHAR(100) NOT NULL,
    Price DECIMAL(10,2) NOT NULL CHECK (Price >= 0),
    StockQty INT NOT NULL DEFAULT 0 CHECK (StockQty >= 0),
    Description VARCHAR(255),

    CONSTRAINT fk_product_seller
        FOREIGN KEY (SellerID)
        REFERENCES Seller(SellerID)
        ON DELETE CASCADE,
```

```
    CONSTAINT fk_product_category
        FOREIGN KEY (CategoryID)
        REFERENCES Category (CategoryID)
    );
-- 6. ORDERS TABLE
--ORDERS is used instead of ORDER because ORDER is
  a MySQL keyword.
CREATE TABLE Orders (
        OrderID VARCHAR (10) PRIMARY KEY,
        CustomerID VARCHAR (10) NOT NULL,
        OrderDate DATE NOT NULL,
        TotalAmt DECIMAL (12,2) NOT NULL CHECK (TotalAmt >=0),
        OrderStatus VARCHAR (20) NOT NULL,

        CONSTRAINT fk_order_customer
            FOREIGN KEY (CustomerID)
            REFERENCES Customer (CustomerID)
            ON DELETE CASCADE
    );
-- 7. ORDER_ITEM TABLE
-- Composite primary key because OrderItem is a weak
  entity.
CREATE TABLE Order_Item (
        OrderID VARCHAR (10) NOT NULL,
        ItemSeq INT NOT NULL,
        ProductID VARCHAR (10) NULL,
        AddressID VARCHAR (10) NULL,
        Quantity INT NOT NULL CHECK (Quantity > 0),
        UnitPrice DECIMAL (10,2) NOT NULL CHECK (UnitPrice >= 0),
        Subtotal DECIMAL (12,2) NOT NULL CHECK (Subtotal >=0),

        PRIMARY KEY (OrderID, ItemSeq),
```

```
    CONSTRAINT fk_orderitem_order
        FOREIGN KEY (Order ID)
        REFERENCES Orders (Order ID)
        ON DELETE CASCADE,
    CONSTRAINT fk_orderitem_product
        FOREIGN KEY (Product ID)
        REFERENCES Product (Product ID)
        ON DELETE SET NULL
);
-- 8, PAYMENT TABLE
CREATE TABLE Payment (
    PID VARCHAR (10) PRIMARY KEY,
    Order ID VARCHAR (10) NOT NULL,
    ItemSeq INT NOT NULL,
    PMode VARCHAR (20) NOT NULL,
    Amt DECIMAL (12,2) NOT NULL CHECK (Amt >= 0),
    PStatus VARCHAR (20) NOT NULL,
    PDate DATE NOT NULL,

    CONSTRAINT uq_payment_order_item
        UNIQUE (OrderID, ItemSq),

    CONSTRAINT fk_payment_orderitem
        FOREIGN KEY (Order ID, Item Seq)
        REFERENCES Order_Item (Order ID, ItemSeq)
        ON DELETE CASCADE

);
```

```
-- 9. DELIVERY TABLE
CREATE TABLE Delivery (
    DID VARCHAR (10) PRIMARY KEY,
    OrderID VARCHAR (10) NOT NULL,
    ItemSeq INT NOT NULL,
    CourierName VARCHAR (50) NOT NULL,
    Status VARCHAR (20) NOT NULL,
    EstDate DATE,
    DeliveryDate DATE,

    CONSTRAINT uq_delivery_order_item
        UNIQUE (OrderID, ItemSeq),

    CONSTRAINT fk_delivery_orderitem
        FOREIGN KEY (OrderID, ItemSeq)
        REFERENCES Order_Item (OrderID, ItemSeq)
        ON DELETE CASCADE

);
-- Display all created tables
SHOW TABLES;
```

SAMPLE Data

```
-- CUSTOMER DATA
INSERT INTO Customer
(CustomerID, FirstName, LastName, Email, DOB, Gender,
 PhoneNo)
VALUES
('C001', 'Krishna', 'Rathore', 'krishna@gmail.com', '2005-05-12',
 'Male' 'XXXXXX7557'),
('C002', 'Vandna', 'Rathore', 'vandna@gmail.com', '2004-08-20',
 'Female', 'XXXXXX777'),
--ADDRESS DATA
INSERT INTO Address
(AddressID, CustomerID, AddressType, IsDefault, Street,
 City, State, Pincode)
VALUES
('A001', 'C001', 'Home', TRUE, 'MG Road', 'Indore', 'M.P.', '452001'),
('A002', 'C002', 'Office', FALSE, 'Vijaya Nagar', 'Indore', 'M.P', '452010'),
('A003', 'C002', 'Home', TRUE, 'Civil Lines', 'Jaipur', Rajasthan, '302006'),
--SELLER DATA
INSERT INTO Seller
(SellerID, SellerName, Phone, Email, GSTIN)
VALUES
('S001', 'Tech World', '9000000001', 'techworld@gmail.com', 'GSTIN01'),
('S002', 'Smart Store', '9000000002', 'smartstore@gmail.com', 'GSTIN002'),
--CATEGORY DATA
INSERT INTO Category (CategoryID, ParentCategoryID)
VALUES
('CAT01', NULL),
('CAT02', 'CAT01'),
('CAT03', 'CAT01');
```

```
-- PRODUCT DATA
INSERT INTO Product
(ProductID, SellerID, CategoryID, ProductName, Price, StockQty,
 Description)
 VALUES
('P001', 'S001', 'CAT02', 'Laptop', 65000.00, 10, 'Core i5 Laptop'),
('P002', 'S001', 'CAT02', 'Smartphone', 30000.00, 20, '5G Smartphone'),
('P003', 'S002', 'CAT03', 'Wireless Mouse', 800.00, 50, 'Wireless Mouse'),
-- ORDER DATA
INSERT INTO Orders
(OrderID, CustomerID, OrderDate, TotalAmt, OrderStatus)
 VALUES
('O001', 'C001', '2026-08-18', 65800.00, 'Confirmed'),
('O002', 'C002', '2026-08-19', 30000.00, 'Shipped'),
-- ORDER ITEM DATA
 INSERT INTO Order_Item
 (OrderID, ItemSeq, ProductID, AddressID, Quantity, UnitPrice,
  Subtotal)
 VALUES
 ('O001', 1, 'P001', 'A001', 1, 65000.00, 65000.00),
 ('O001', 2, 'P003', 'A001', 1, 800.00, 800.00),
 ('O002', 1, "P002", 'A003', 1, 30000.00, 30000.00);
 -- PAYMENT DATA
 INSERT INTO Payment
 (PID, OrderID, ItemSeq, PMode, Amt, PStatus, PDate)
 VALUES
 ('PM001', 'O001', 1, 'UPI', 65000.00, 'Paid', '2026-08-18'),
 ('PM002', 'O001', 2, 'UPI', 800.00, 'Paid', '2026-08-18'),
 ('PM003', 'O002', 1, 'Card', 30000.00, 'Paid', '2026-08-19');
```

-- DELIVERY DATA

```
INSERT INTO Delivery
(DID, OrderID, ItemSeq, CourierName, Status, EstDate,
DeliveryDate).
VALUES
('D001', '0001', 1, 'BlueDart', 'Delivered', '2026-08-20', '2026-08-20'),
('D002', '0001', 2, 'Delhivery', 'Delivered', '2026-08-20', '2026-08-20'),
('D003', '0002', 1, 'EcomExpress', 'In Transit', '2026-08-22', NULL);
```

Output

1. Display Tables

```
SHOW TABLES;
```

Output:

Tables_in_dbms_lab
address
category
customer
delivery
order_item
orders
payment
product
seller

9 rows in set

2. Display Customer Data

```
SELECT CustomerID, FirstName, LastName, Email, PhoneNo
FROM Customer;
```

Output:

CustomerID	First Name	LastName	Email	Phone No.
C001	Krishna	Rathore	k@gmail.com	xxxxxx0000
C002	Vedika	Patel	v@gmail.com	xxxxxx0001

2 rows in set

3. Display Orders with Customer Names

```
SELECT O.OrderID, CONCAT(C.FirstName, ' ', C.LastName) AS
       CustomerName, O.TotalAmt, O.OrderStatus
FROM Orders O
JOIN Customer C ON O.CustomerID = C.CustomerID;
```

Output:

OrderID	Customer Name	TotalAmt	OrderStatus
0001	Krishna Rathore	65800.00	Confirmed
0002	Vandna Rathore	30000.00	Shipped

2 rows in set

4. Display Order Items

```
SELECT OI.OrderID, OI.ItemSeq, P.ProductName,
       OI.Quantity, OI.UnitPrice, OI.Subtotal
FROM Order_Item OI
LEFT JOIN Product P ON OI.ProductID = P.ProductID;
```

Output:

OrderID	ItemSeq	ProductName	Quantity	UnitPrice	Subtotal
0001	1	Laptop	1	65000.00	65000.00
0001	2	Wireless Mouse	1	800.00	800.00
0002	1	SmartPhone	1	30000.00	30000.00

3 rows in set

5. Foreign Key Violation

The following statement tries to insert an order item for a non-existing order 0999.

```
INSERT INTO Order_Item
(OrderID, ItemSeq, ProductID, AddressID, Quantity, UnitPrice,
 Subtotal)
VALUE
('0999', 1, 'P001', 'A001', 1, 65000.00, 65000.00);
```

Output:

ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails

Reason: OrderID = '9999' does not exist in the Orders table.

6. Another Foreign Key Violation

```
INSERT INTO Product
(ProductID, SellerID, CategoryID, ProductName, Price,
 StockQty, Description)
VALUES
('P999', 'S999', 'CAT02', 'Test Product', 1000.00, 5,
 'Invalid seller');
```

Output:

ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails

Reason: SellerID = 'S999' does not exist in the Seller table

ON DELETE SET NULL Demonstration:

```
DELETE FROM Category
WHERE CategoryID = 'CAT01';

SELECT CategoryID, ParentCategoryID
FROM Category;
```

Output:

CategoryID	ParentCategoryID
CAT02	NULL
CAT03	NULL

The child categories remain, but their ParentCategoryID become NULL.

ON DELETE CASCADE Demonstration

```
DELETE FROM Customers
WHERE CustomerID = 'C001';
```

Because of ON DELETE CASCADE, the following related records are automatically deleted:

Customer C001
↓
Addresses A001, A002
↓
Orders O001
↓
Order Items O001-1, O001-2
↓
Related Payments
↓
Related Deliveries

Check the result:

```
SELECT * FROM Orders
WHERE CustomerID = 'C001';
```

Output: Empty Set

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